DATE: DAY:

5/11/25 (A.P.T)

:steps
(i) Penalities calculations (Two smallest values difference)
(ii) Maximum Choose from Penality
(iii) Least cost from

TP (Transportation Problem)
(i) NWM
(ii) Least M

⑥ Vogel's Approximation Method:

→ VAM (Penalty Method)

⊙ Vogel's approximation method is another method to find out initial feasible solution of transportation problems. VAM is a heuristic (optimal)

→ **Ex:-**

The following transportation tableau summarizes the supply, demand and cost of transportation (in Rs) for five factories (1 to 5) producing goods for five warehouses (A through e)

⋮

Find the initial feasible solution using vogel's approximation method.

Sol:- Calculations of Penalities

	A	B	C	D	E	supply		
1	4 (25)	11	17	12	11	~~17~~ 0	25	
2	~~3~~	6 (12)	15 (12)	19	11 (15)	~~32~~ ~~25~~	~~7~~ ~~2~~	
3	7	9	10	7	14	19		
4	10	9	9 (11)	15	16	~~12~~ 0		
5	12	6	7 (14)	10 (16)	19	20		
demand	~~24~~ 7 6	~~18~~ 0	~~26~~ 16 14	25	~~5~~ 0	100		

	P_1	P_2	P_3	P_4	P_5	P_6
P_1	1	2	2	3	0	
P_2	4	2	2	3	3	
P_3	–	2	2	3	3	
P_4	–	–	3	3	3	
P_5	–	–	2	3	3	
P_6	–	–	–	3	–	

P_1	P_2	P_3	P_4	P_5	P_6
11-4 = 7 ← Maximum	–	–	–	–	
8 - 3 = 5	(5)	(3)	4	(4)	(4)
~~7~~ - 7 = 0	0	2	3	3	3
~~7~~ 9 - 9 = 0	0	0	(6)	–	–
7 - 6 = 1		1	3	3	3

• Number of allocated cells = m + n - 1
= 5 + 5 - 1
= 9

• Minimum cost of transportation = 4×17 + 3×7 + 8×8 + 15×2 + 11×5 + 9×12 + 7×14 + 10×6

DATE: DAY:

→ Transport from :

- Factory 1 to warehouses A
- Factory 2 to A, B, C, D, E
- Factory 3 to D
- Factory 4 to C
- Factory 5 to C and D

Example : 6

From \ To	1	2	3	4	Supply
A	12	4	9	5	55
B	8	1	6	6	45
C	1	12	84	7	30
Demand	40	20	50	20	

DATE DAY:

10/11/25

* Assignment Problem (Hungarian Method):

→ Assign jobs to a person to minimize costs

Jobs	Person/Machine
1	P1
2	P2
3	P3
4	P4
5	P5

: No. of Jobs = Persons/Machines (Balanced Problem)

→ Suppose there are 'n' jobs to be performed and 'n' persons are available for doing jobs. Assume that each person can do each at a time. let c_{ij} be the cost of ith person is assigned the jth job. The problem is that assigned the job, so that total cost is minimized of performing all the jobs.

DATE: _____ DAY: _____

Q: Solve the AP by using Hungarian Method. Assign the jobs for different machines. So as to minimize the total cost.

Job / Machine	A	B	C	D	E
1	13	8	16	18	19
2	9	15	24	9	12
3	12	9	4	4	4
4	6	12	10	8	13
5	15	17	18	12	20

: We treat as a Matrix

Step1:

⇒ Since we are treating above table as Matrix and number of rows is equal to number of columns, so problem is balanced.

⊙ Step2: Row Reduction Matrix:

→ subtract the minimum element from all the elements in respective row. We get the row reduction Matrix.

DATE: DAY:

Jobs/ Machines	A	B	C	D	E
1	5	0	8	10	11
2	0	6	15	0	3
3	8	5	0	0	0
4	0	6	4	2	7
5	3	5	6	0	8

⊙ Step 3: Column Reduction Matrix

→ Select the minimum element from all the elements in respective column we get the column reduction Matrix.

Jobs/ Machines	A	B	C	D	E
1	5	0	8	10	11
2	0	6	15	0	3
3	8	5	0	0	0
4	0	6	4	2	7
5	3	5	6	0	8

⊙ Step 4: Draw maximum Horizontal and Vertical line(N) to cover all zeros.

: Max 1 zero in

DATE: _____ DAY: _____

Jobs/machines	A	B	C	D	E
1	5	9	8	10	11
2.	4	6	15	9	3
3	8	5	9	9	9
4	9	6	11	2	7
5	3	5	6	1	8

: Max 1 zero in raw or column

a) No of N = n (order of M)
Then it is optimal.

b) No of N ≠ n
Go to next step

: As in above table
No. of N ≠ n
3 ≠ 5 (so Next Step)

⊙ Step 5: Modified Matrix

(i) Determine smallest Uncover (No horizontal and vertical line) number (value) = x = 1

: Subtract 1 from each element in Matrix.

Jobs/Machines					

DATE: DAY:

Q: Solve the assignment model

Jobs	1	2	3	4	5	6
A	9	22	58	11	19	27
B	43	78	72	50	63	48
C	41	28	91	37	45	33
D	74	42	27	49	39	32
E	36	11	57	22	25	18
F	3	56	53	31	17	28

⊙ Step 1:

Since the problem is balanced as no. of rows is equal to no. of columns.

⊙ Step 2: Row Reduction

	1	2	3	4	5	6
A	0	13	49	2	10	18
B	0	35	28	7	20	5
C	13	0	63	9	17	5
D	47	15	0	22	12	5
E	25	0	46	11	14	7
F	0	53	50	28	14	25

DATE. DAY.

⊙ Step 3: Column Reduction

	1	2	3	4	5	6
A	0	13	49	0	0	13
B	0	35	29	5	10	0
C	13	0	63	7	7	0
D	47	15	0	20	2	0
E	25	10	46	9	4	2
F	0	53	50	26	4	20

12/11/25

⊙ Solve the AP by using hungarian method. Assign the jobs for different machines. so as to minimize the total cost.

Assignment Problem
(One to one correspondence)

Balance ↙ ↓ Unbalance ↘ Travel

Jobs	A	B	C	D	E
1	13	8	16	18	19
2	9	15	24	9	12
3	12	9	4	4	4
4	6	12	10	8	13
5	15	17	18	12	20

DATE: DAY:

Sol:-

Since No. of rows = No. of columns

=> Given problem is balanced

→ **step1**: Row Reduction Matrix

Job/M	A	B	C	D	E
1	5	0	8	10	11
2	0	6	15	0	3
3	8	5	0	0	0
4	0	6	4	2	7
5	3	5	6	0	8

→ **step2**: Column Reduction Matrix

Job/M	A	B	C	D	E
1	5	0	8	10	11
2	0	6	15	0	3
3	8	5	0	0	0
4	0	6	4	2	7
5	3	5	6	0	8

: Now draw horizontal and vertical lines to cover one zero row.

DATE: : Scan row & draw on column : First select Horizontal then vertical DAY: _____

→ **Step 3:** Draw minimum Horizontal and vertical Line to cover all zeros.

Job/M	A	B	C	D	E
1	5	0	8 -3	10	11 -3
2	0	6	15 -3	0	3 -3 smallest
3	8 +3	5 +3	0	0 +3	0
4	0	6	4 -3	2	7 -3
5	3	5	6 -3	0	8 -3

(a) No of line N = n (Order Matrix)
Solution is optimal

(b) N < n
Go to next step (Modified Matrix)

→ In above 4 < 5 (2nd condition)
Now Determine the smallest uncover element = x = 3

→ Now write Uncover values - x
Intersection values + x
: Write all other values same.

DATE: DAY:

→ Modified Matrix (Add and subtract minimum)

Jobs / M	A	B	C	D	E
1	5	0	5	10	8
2	0	6	12	0	0
3	1	9	0	3	0
4	0	6	1	2	4
5	3	5	3	0	5

: N=5=n

Now, it is optimal solution.

⇒ So we can assign the jobs to the machines.

Jobs	Machines
1	B
2	E
3	C
4	A
5	D

DATE: _____ DAY: _____

Q:

Jobs/Persons	P_1	P_2	P_3	P_4
J_1	5	7	11	6
J_2	8	5	9	6
J_3	4	7	10	7
J_4	10	4	8	3

Sol:-

→ Since No. of raws = No. of columns

⇒ Given problem is balanced

→ Step 1: Row - Reduction Matrix.

Jobs/Persons	P_1	P_2	P_3	P_4
J_1	0	2	6	1
J_2	3	0	4	1
J_3	0	3	6	3
J_4	7	1	5	0

→ Step 2: Column - Reduction Matrix

Jobs/Persons	P_1	P_2	P_3	P_4
J_1	0	2	2	1
J_2	3	0	0	1
J_3	0	3	2	3
J_4	7	1	1	0

DATE: DAY:

⊙ Step 3: Draw minimum Horizontal and vertical lines to cover all the zeros

Jobs/Persons	P_1	P_2	P_3	P_4
J_1	[0]	2	2	1
J_2	~~3~~	[0]	0	~~1~~
J_3	0	3	2	3
J_4	~~7~~	1	1	[0]

$N < n$

$3 < 4$ Go to next step- : Minimum = 1

⊙ Modified Matrix:

Jobs/Persons	P_1	P_2	P_3	P_4
J_1	[0]	1	1	1
J_2	~~1~~	0	0	~~2~~
J_3	0	2	1	3
J_4	~~7~~	0	0	[0]

$2 < 4$: Minimum = 1

⇒ Uncover - Minimum

Intersect + Minimum

: Delete zero row with maximum zeros

DATE: _____ DAY: _____

Jobs/Persons	P_1	P_2	P_3	P_4
J_1	0	0	0	0
J_2	5	0	0	2
J_3	0	1	0	2
J_4	9	0	0	0

: Del rows with maximum to minimum ~~zeros~~

$N=4$ & $n=4$

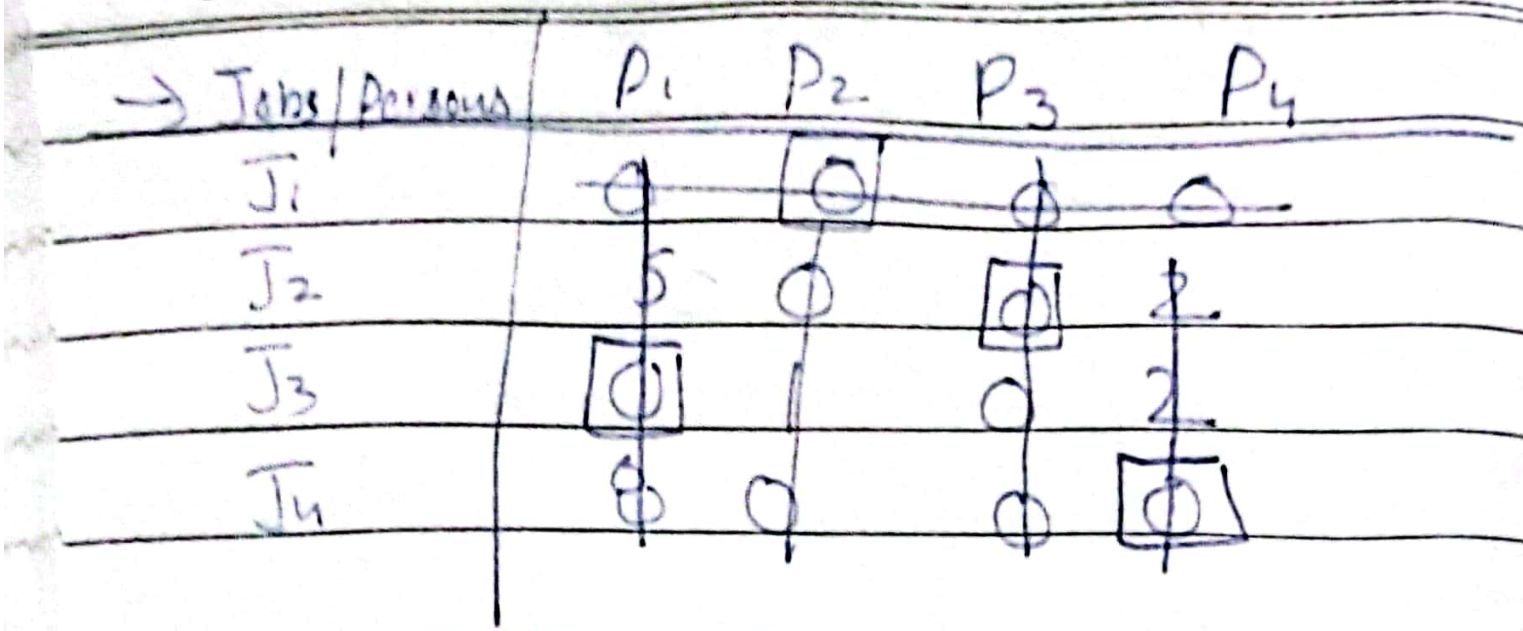
⇒ It is optimal but we can't draw horizontal and vertical lines now, so we do randomly.

Jobs/Persons	P	P_2	P_3	P_4
J_1	0	0	0	0
J_2	5	0	0	2
J_3	0	1	0	2
J_4	9	0	0	0

→ Now Assign jobs to person:

Jobs	Persons	Minimum Cost
J_1	P_1	= 5+5+10+3
J_2	P_2	=23
J_3	P_3	
J_4	P_4	

DATE: DAY:



$J_1 \longrightarrow P_2$

$J_2 \longrightarrow P_3$

$J_3 \longrightarrow P_1$

$J_4 \longrightarrow P_4$

DATE: DAY:

19/11/25

(+) Quiz #3 (After Mid) on Wednesday.

(+) Network Analysis:

→ Real Life Examples:

: ⊙ Building Paint (order matters)

⊙⊙ Foundation ② Walls

③ Install roof

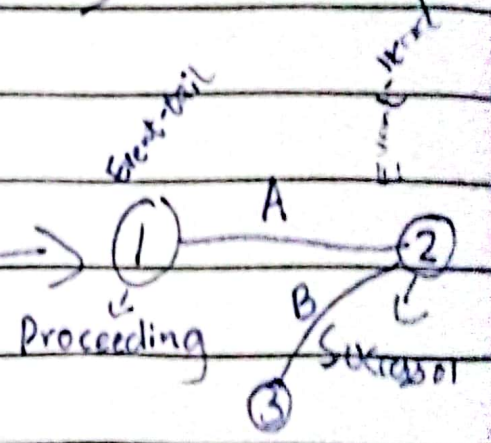
④ Doors and windows

: ⊙ Wedding.

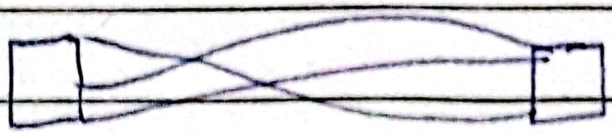
Book Hall → ....

→ **Network:**

① Activity →



(+) Dijkstra's Algorithm:



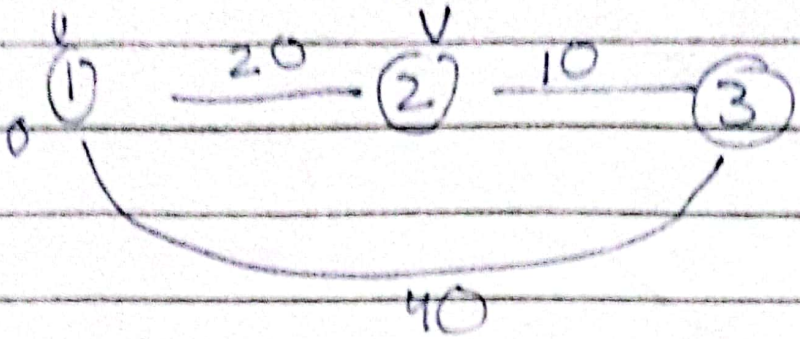
→ Using network moving from one node to other node.

→ we use it when cost and distance is positive (maximum)

DATE: DAY:

→ E.g: Shortest distance, cheapest path, Fast processing of computers, Google Maps

→



* Select source (E.g: ①)
* Select Destination (E.g: ②, ③)

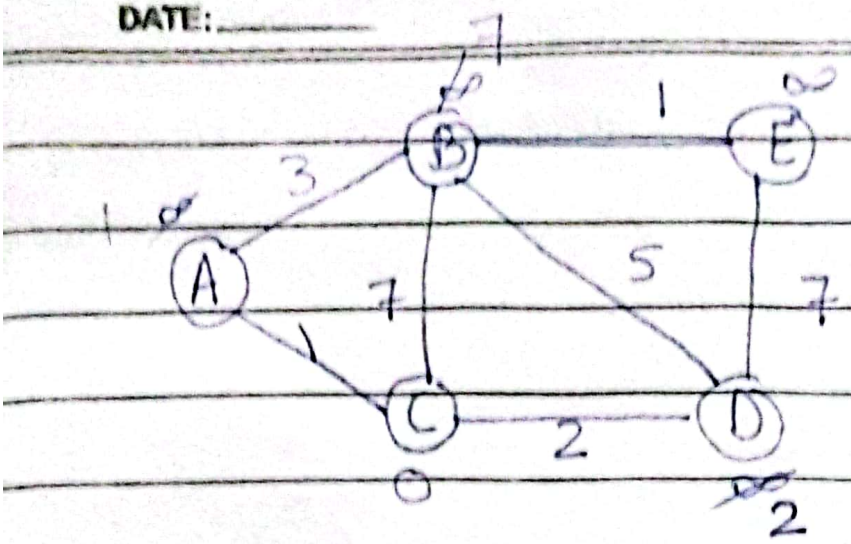
1 → 3 (cost = 40)
1 → 2 → 3 (cost = 30)

→ Relaxation:

If $d(u) + ((u,v) < dv)$
then $d(0) + ((u,v) = d(v))$
E.g: If $0+20 < \infty$
then $20 = d(v)$

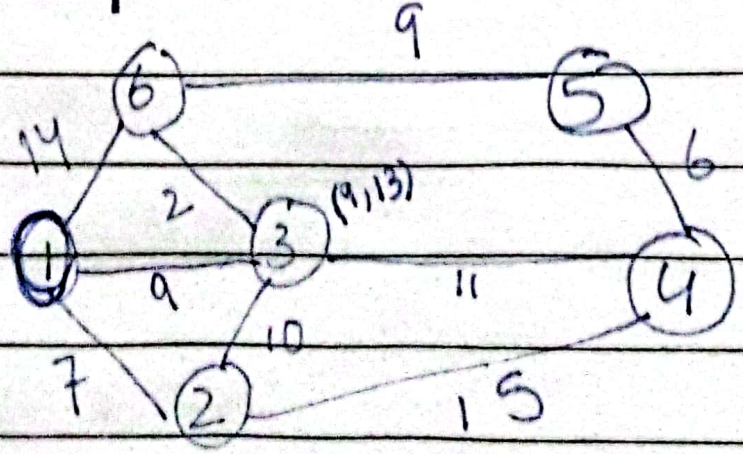
Ex Find the shortest path in given weight graph by Dijkstra's algorithm

DATE: DAY:



Source C	Destination A	B	D	E
	∞	∞	∞	∞
C, A	(1)	7	2	∞
C, A, D	-	4	(2)	∞
C, A, D, B	-	(4)	-	9
C, A, D, B	-	-	-	(5)

⑨ Example:

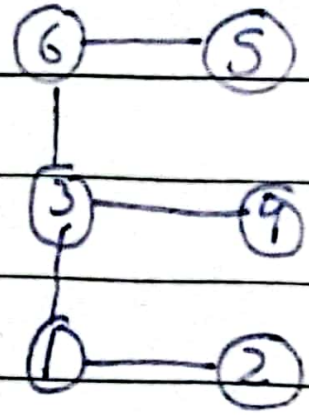


DATE: _____ DAY: _____

Source 1	2	3	4	5	6
	∞	∞	∞	∞	∞
1,2	(7)	9	∞	∞	14
1,2,3	-	(9)	22	∞	14
1,2,3,6	-	-	20	∞	(11)
1,2,3,6,4	-	-	(20)	20	-
1,2,3,6,4,5	-	-	-	(20)	-

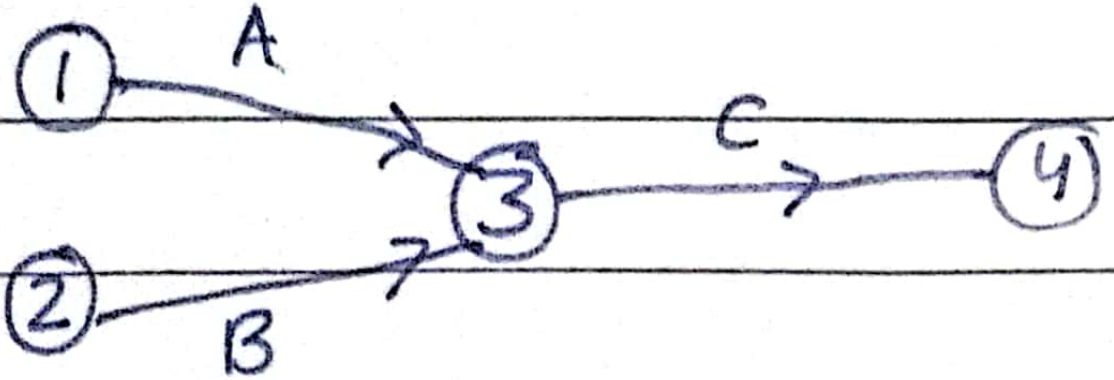
→

	Dis
1 → 2	7
1 → 3	9
1 → 3 → 4	20
1 → 3 → 6 → 5	20

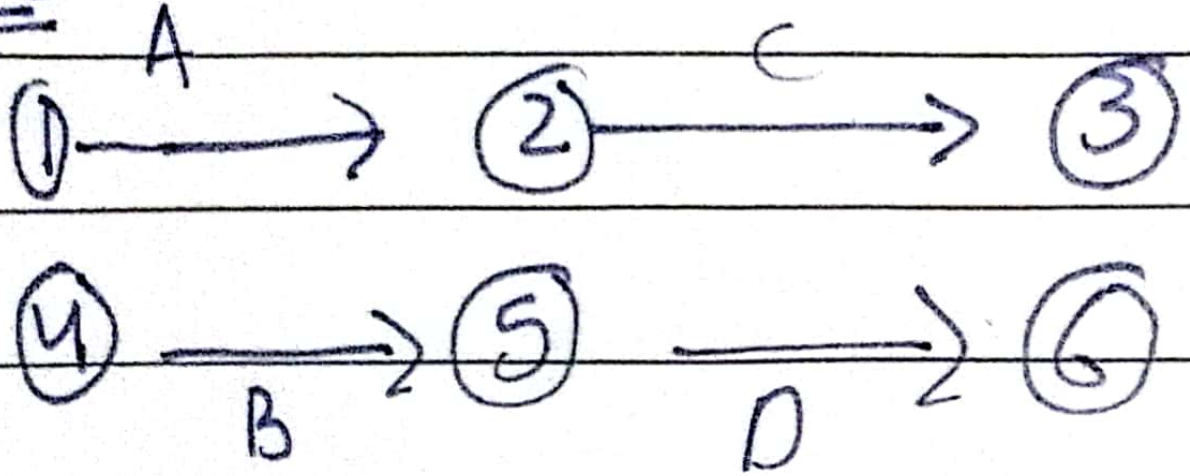


24/11/25

* Ex: 1



* Ex: 2

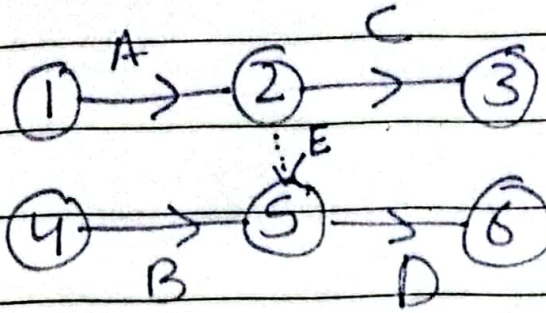


DATE: DAY:

Ex:1 (*) Activity ← Nodes

Ex#1 (•) The activities A and B can start simultaneously, but activity C starts after the completion of both A and B.

Ex#2:

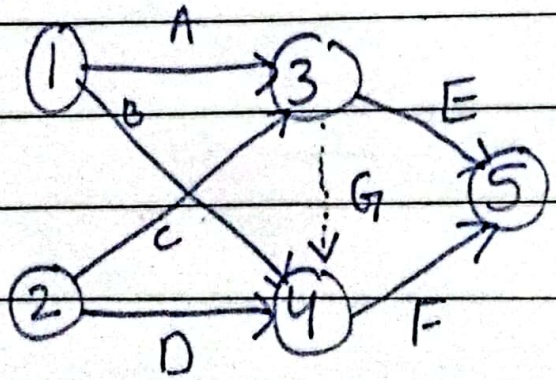


: Not connected network

: Dotted Line (Dummy connection Activity)

→ The activities A and B can be started simultaneously, C starts after A is completed, but D must wait till activities A and B are completed.

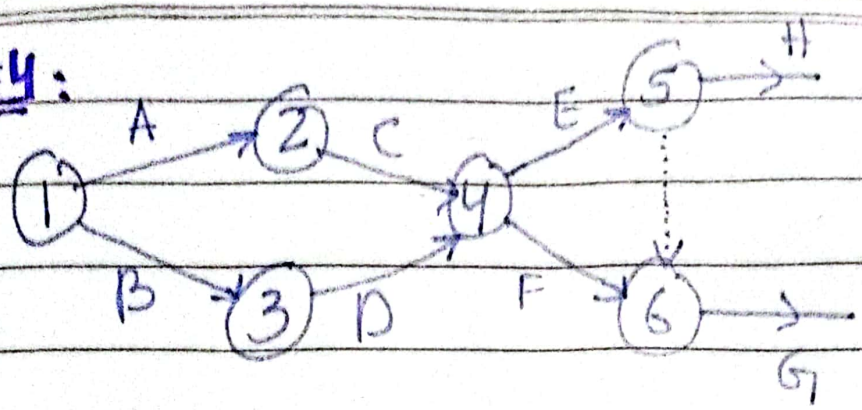
Ex#3:



→ The activities A, B, C, D can be started simultaneously .....

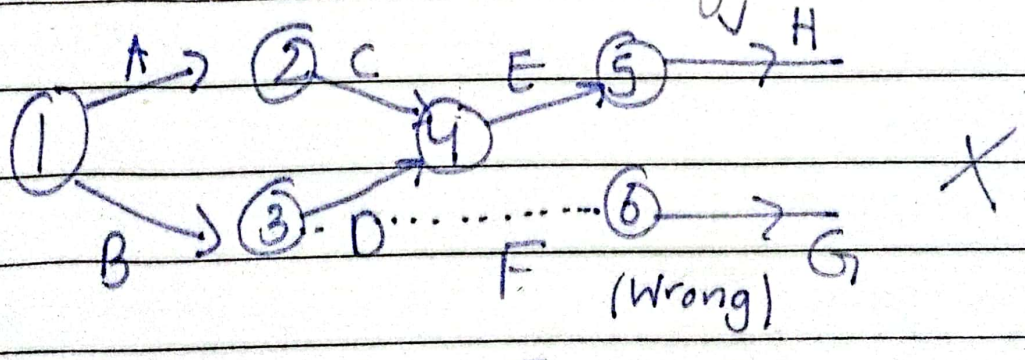
DATE: DAY:

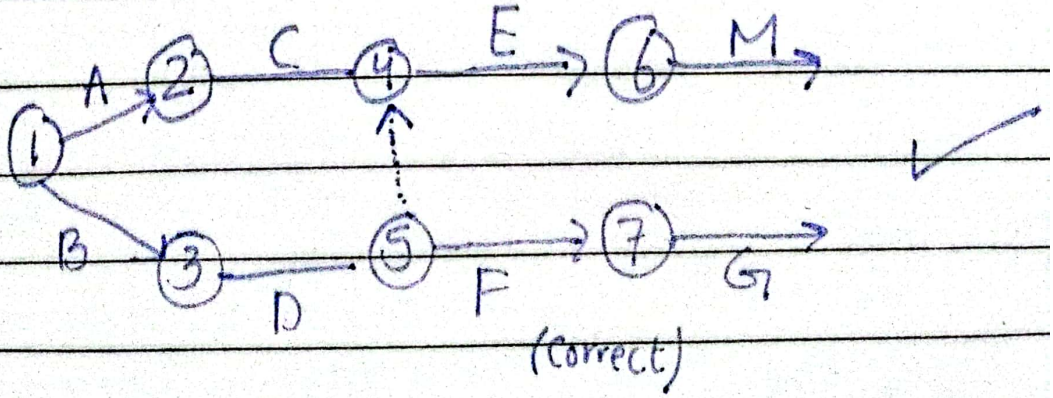
Ex #4:

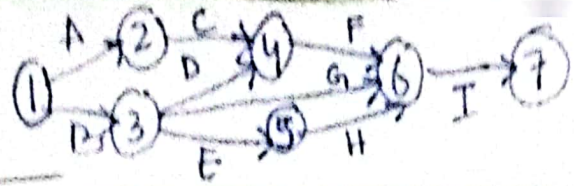


→ A and B start simultaneously, C, depend on A and D depend on B .....

Q: If we want activity E depends only only on the completion of both activities C and D and F depands on D alone - Write modify network.







DATE: DAY:

Activity	Predecessor	Successors
A	—	C
B	—	D, G, E
C	A	F
D	B	F
E	B	H
F	C, D	I
G	B	I
H	C, E	I
I	F, G, H	—

* Ex:5

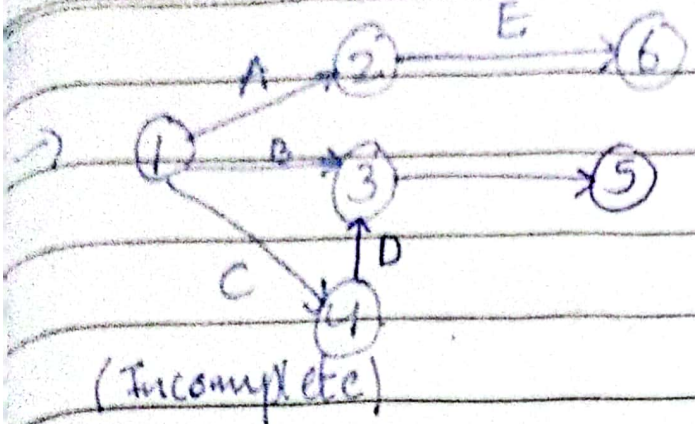
Draw the network for the following Project

Activity	Predecessor
A	—
B	—
C	—
D	C
E	A
F	B, D
G	B, D
H	C
I	E, F, G
J	H, I

: Last one not present in predecessor

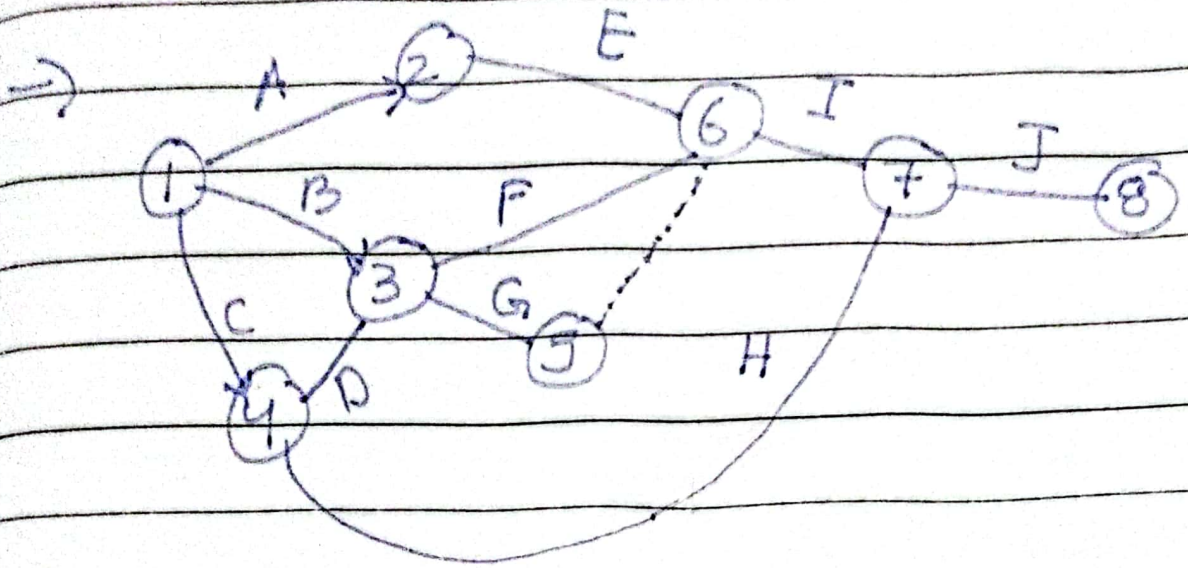
: Merge with two predecessor

DATE: DAY:



(Incomplete)

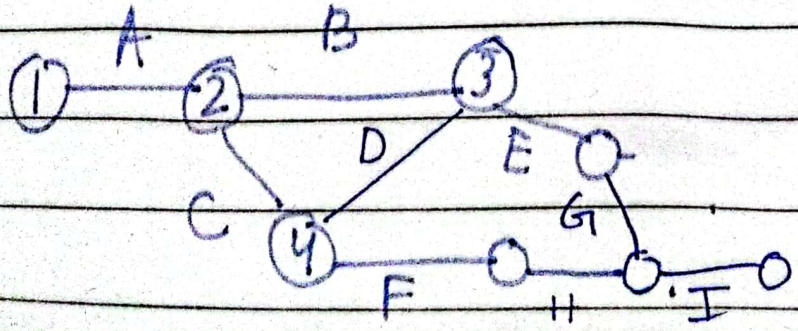
: Ex: 10.6
(Attempt it)



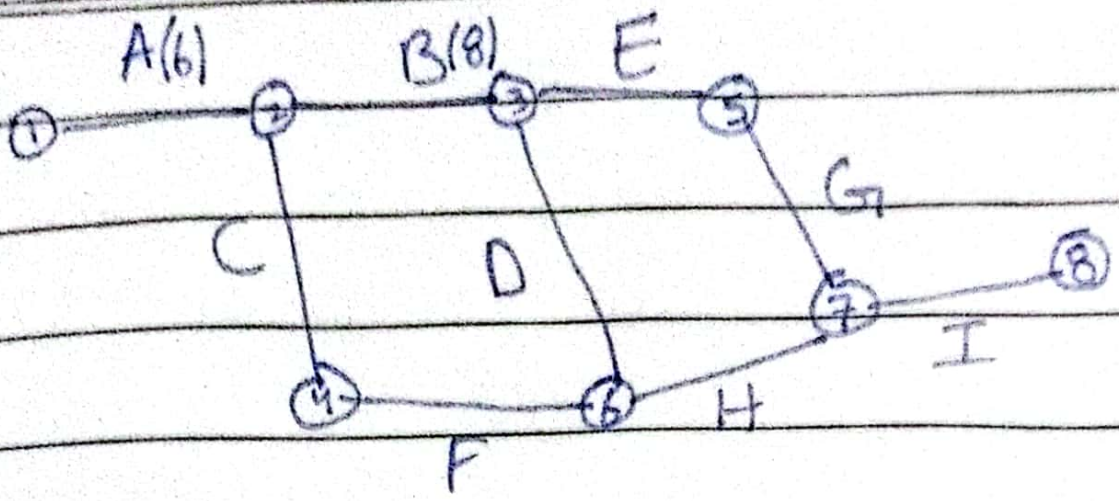
26/11/25

Draw the diagram from the following activities

Activities	A	B	C	D	E	F	G	H	I
Predecessor	-	A	A	B	B	C	E	D,F	G,H
Time	6	8	8	20	18	10	8	2	7



DATE: DAY:



(4)

Activities	A	B	C	D	E	F	G	H	I	J
Immediate Predecessor	-	-	A	A	I, J, K	B, D	B, D	F	A	G,

